Liangyu WU | Curriculum Vitae

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Education Background

Stanford University **California, U.S.**

Ph.D. Candidate in Physics, Physics Department (Advisor: **Dr. Julia Gonski** & **Dr. Dong Su**) Sep. 2024-Present

- Research area: Experimental Particle Physics

Shanghai Jiao Tong University **Shanghai, China**

Bachelor of Science in Physics, School of Physics and Astronomy Sep. 2020- Jun. 2024

- Coursework: *Quantum Mechanics*, *Electrodynamics*, *Equations of Mathematical Physics*, *Professional Experiment*, *Introduction to Nuclear and Particle Physics*, *Methods of Experimental Nuclear and Particle Physics* (All graded A's)

University of Maryland, College Park **Maryland, U.S.**

Visiting International Student Aug. 2023-Dec. 2023

Research Experience

SLAC National Accelerator Laboratory **California, U.S.**

Particle Physics and Astrophysics Jan. 2025-Present

Research Assistant (Supervisor: **Dr. Julia Gonski** & **Dr. Ariel Schwartzman** & **Dr. Dong Su**)

– **Research on the Future Colliders**

- Development of real-time machine learning algorithms for deployment on front-end electronics in future detectors.
- Agentic AI-driven optimization of detector design and reconstruction algorithms.
- R&D of embedded FPGA (eFPGA) technology for ultra-low latency signal processing applications.
- Investigation of novel digital sensor technologies (e.g., digital SiPMs) for calorimetry readout.

– **Research on the ATLAS Experiment at CERN**

- Anomaly-detection searches for new-physics signatures with data-driven, model-agnostic techniques.
- Development of novel foundation models for collider physics and new physics searches.
- Testing of custom front-end ASICs for high-speed data transmission in the ATLAS ITk upgrade for HL-LHC.
- Advancing LAr calorimeter timing to extend fast-timing detector coverage beyond HGTD.

SLAC National Accelerator Laboratory **California, U.S.**

Particle Physics and Astrophysics Sep. 2024-Dec. 2024

Research Assistant (Supervisor: **Prof. Spencer Gessner**)

– **Research on the FACET-II**

- Development of user interface software and data analysis tools for advanced accelerator systems.

University of Maryland, College Park **Maryland, U.S.**

Department of Physics Aug. 2023-Aug. 2024

Undergraduate Research Student (Supervisor: **Prof. Christopher Palmer** & **Prof. Sarah Eno**)

– **Research on the Dual-Readout Calorimetry**

- Monte Carlo simulation studies of detector response optimization across diverse calorimeters.
- Mathematical modeling and algorithm development for multi-channel signal processing and calibration.

Shanghai Jiao Tong University **Shanghai, China**

School of Physics and Astronomy Oct. 2021-Feb. 2024

Undergraduate Research Assistant (Supervisor: **Prof. Yue Meng**)

– **R&D for a Novel Radon Detector**

- Design and construction of novel detector with precision measurement capabilities for material characterization.

PandaX Collaboration **Shanghai & Sichuan, China**

Undergraduate Research Student Oct. 2021-Feb. 2023

– Work in Ultra-low background technique R&D group

- Material characterization and ultra-low background technique implementation for precision detection systems.

– Work in PMT group

- Testing, optimization, and installation of high-voltage detection systems and power distribution infrastructure.

Tsung-Dao Lee Institute **Shanghai, China**

Astronomy and Astrophysics Division Mar. 2023-Aug. 2023

Undergraduate Research Student (Supervisor: **Prof. Masahiro Ogihara**)

– Research on the stability of unevenly spaced planetary systems

- N-body computational simulations for dynamic system stability analysis.
- Statistical modeling demonstrating systematic biases in conventional analytical approaches.

Publications

Instrumentation & Detector R&D

[1] **Towards Foundation Models on Hardware Accelerators for Particle Physics**, *M. Benyas, Q. Liu, V. Mikuni, J. Gonski, P. A. May, B. Nachman, T. Wamorkar,* ***L. Wu***, ML4PS Workshop 2026, arXiv:2609.26899

[2] **Leveraging Industrial Foundation Models at the Edge of Particle Physics Detectors via Distillation Learning and Hardware Co-design**, *G. Ancone, Q. Liu,* ***L. Wu****, J. Gonski*, ML4PS Workshop 2026, arXiv:2609.23385

[3] **Machine Learning Enables Real-Time Waveform Decomposition for Dual-Readout Calorimetry**, ***L. Wu****, Q. Liu, M. T. Lucchini, J. Gonski, M. Campajola, S. Moneta*, arXiv:2604.26090

[4] **Agentic-AI Detector Co-design and Optimization in Vertically-Integrated Differentiable Full Simulations**, *W. Chung, Q. Liu,* ***L. Wu****, J. Gonski*, Accepted to Stanford Physics & AI Conference 2026

[5] **Machine Learning on Heterogeneous, Edge, and Quantum Hardware for Particle Physics**, *J. Gonski, J. Ott, S. Abbaszadeh, S. Addepalli, M. Cremonesi et al.*, PRX Intell. 1, 021201 (2026)

[6] **Edge Machine Learning for Cluster Counting in Next-Generation Drift Chambers**, *D. Yilmaz,* ***L. Wu*** *et al.*, Accepted by NeurIPS 2025

[7] **On the resolution of dual readout calorimeters**, *S. Eno,* ***L. Wu*** *et al.*, NIM A, Volume 1083, 2026, 171080, ISSN 0168-9002

[8] **Design and Experimental Application of a Radon Diffusion Chamber for Determining Diffusion Coefficients in Membrane Materials**, ***L. Wu****, L. Si et al.*, JINST, 20, no.03, P03031 (2025)

[9] **Geant4 simulations of sampling and homogeneous hadronic calorimeters with dual readout for future colliders**, *S.V. Chekanov, S. Eno, S. Magill, C. Palmer,* ***L. Wu***, NIM A, Volume 1072, 2025, 170200, ISSN 0168-9002

Physics Analysis

[1] **Towards anomaly detection searches for new physics signatures including Higgs bosons with weakly supervised machine learning**, *C. L. Cheng, J. Gonski, R. Li, Q. Liu, B. Nachman, D. Noll, J. K. Romman,* ***L. Wu***, Submitted to JHEP

Other

[1] **A Lightweight Foundation Model for Collider Physics with Multi-Domain Adaptation**, ***L. Wu****, Q. Liu, A. Yue, J. Gonski*, arXiv:2607.27501

[2] **The stability of unevenly spaced planetary systems**, *S. Yang,* ***L. Wu*** *et al.*, ICARUS, Volume 406, December 2023

Selected Honors and Awards

2026 Early Career Professional Award, Future Circular Collider (FCC) Week

2024 Shanghai Outstanding Graduate honors, Shanghai Education Commission

Conference Talks & Posters

Conference Talks

1. [Mar 2026] **ITk Week 2026**
   - *GBCR Testing*

2. [Dec 2025] **ATLAS Timing Workshop**
   - *Vertex t0 Reconstruction Using LAr Calorimeter with Outlook for LAr-HGTD Combination*
3. [Oct 2025] **Coordinating Panel on Advanced Detectors (CPAD) 2025**
   - *Developing Digital Silicon Photomultiplier (dSiPM) Specifications for a High-Granularity Calorimeter with Simulations*
   - *Embedded ML Solutions for Real-time Processing in Future Drift Chambers*
4. [Jul 2025] **US ATLAS Summer Workshop 2025**
   - *Enhancing Timing Information Reconstruction with LAr Calorimeter*
5. [May 2025] **ITk Week 2025**
   - *Recent Results of GBCR3 Testing at SLAC*

Posters

1. [Jun 2026] **ATLAS Week 2026**
   - *The GBCR ASIC for ATLAS ITk Pixel Readout: Testing Framework and Performance Validation*
2. [Jun 2026] **FCC Week 2026**
   - *Machine Learning Enables Real-Time Waveform Decomposition for Dual-Readout Calorimetry*

Skills

Technical: C++, Python, CERN ROOT, Geant4, SolidWorks, COMSOL, LaTeX

Hardware: Silicon microelectronics, Printed circuit boards (I/O, transmission, debugging), High-voltage technology

Language: English (Fluent), Mandarin (Native)

Extra-Curricular Experience

Student Union of School of Physics and Astronomy **Shanghai, China**

Co-President of the Student Union Dec. 2021-Dec. 2022

– **Spearheaded the execution of events and promotions for the Student Union.**